GROUP1 SPRINT / FOUNDATION READING

Proportions and their applications

General Mental Ability / English

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6.5 Solve Proportions and their Applications

Learning Objectives

By the end of this section, you will be able to:

- Use the definition of proportion
- Solve proportions
- Solve applications using proportions
- Write percent equations as proportions
- Translate and solve percent proportions

Be Prepared!

Before you get started, take this readiness quiz.

1. Simplify: $\frac{\frac{1}{3}}{4}$.
   If you missed this problem, review **Example 4.44**.
2. Solve: $\frac{x}{4} = 20$.
   If you missed this problem, review **Example 4.99**.
3. Write as a rate: Sale rode his bike 24 miles in 2 hours.
   If you missed this problem, review **Example 5.63**.

Use the Definition of Proportion

In the section on Ratios and Rates we saw some ways they are used in our daily lives. When two ratios or rates are equal, the equation relating them is called a **proportion**.

Proportion

A proportion is an equation of the form $\frac{a}{b} = \frac{c}{d}$, where $b \neq 0, d \neq 0$.

The proportion states two ratios or rates are equal. The proportion is read "a is to b, as c is to d".

The equation $\frac{1}{2} = \frac{4}{8}$ is a proportion because the two fractions are equal. The proportion $\frac{1}{2} = \frac{4}{8}$ is read "1 is to 2 as 4 is to 8".

If we compare quantities with units, we have to be sure we are comparing them in the right order. For example, in the proportion $\frac{20 \text{ students}}{1 \text{ teacher}} = \frac{60 \text{ students}}{3 \text{ teachers}}$ we compare the number of students to the number of teachers. We put students in the numerators and teachers in the denominators.

EXAMPLE 6.40

Write each sentence as a proportion:

ⓐ 3 is to 7 as 15 is to 35. ⓑ 5 hits in 8 at bats is the same as 30 hits in 48 at-bats.

ⓒ $\$1.50$ for 6 ounces is equivalent to $\$2.25$ for 9 ounces.

Solution

ⓐ

	3 is to 7 as 15 is to 35.
Write as a proportion.	$\frac{3}{7}=\frac{15}{35}$

ⓑ

	5 hits in 8 at-bats is the same as 30 hits in 48 at-bats.
Write each fraction to compare hits to at-bats.	$\frac{\text{hits}}{\text{at-bats}}=\frac{\text{hits}}{\text{at-bats}}$
Write as a proportion.	$\frac{5}{8}=\frac{30}{48}$

ⓒ

	$1.50 for 6 ounces is equivalent to $2.25 for 9 ounces.
Write each fraction to compare dollars to ounces.	$\frac{\$}{\text{ounces}}=\frac{\$}{\text{ounces}}$
Write as a proportion.	$\frac{1.50}{6}=\frac{2.25}{9}$

TRY IT : : 6.79

Write each sentence as a proportion:

ⓐ 5 is to 9 as 20 is to 36. ⓑ 7 hits in 11 at-bats is the same as 28 hits in 44 at-bats.

ⓒ $\$2.50$ for 8 ounces is equivalent to $\$3.75$ for 12 ounces.

TRY IT : : 6.80

Write each sentence as a proportion:

ⓐ 6 is to 7 as 36 is to 42. ⓑ 8 adults for 36 children is the same as 12 adults for 54 children.

ⓒ $\$3.75$ for 6 ounces is equivalent to $\$2.50$ for 4 ounces.

Look at the proportions $\frac{1}{2}=\frac{4}{8}$ and $\frac{2}{3}=\frac{6}{9}$. From our work with equivalent fractions we know these equations are true.

But how do we know if an equation is a proportion with equivalent fractions if it contains fractions with larger numbers?

To determine if a proportion is true, we find the **cross products** of each proportion. To find the cross products, we multiply each denominator with the opposite numerator (diagonally across the equal sign). The results are called a cross products because of the cross formed. The cross products of a proportion are equal.

$8 \cdot 1 = 8 \quad 2 \cdot 4 = 8 \qquad 9 \cdot 2 = 18 \quad 3 \cdot 6 = 18$

$\frac{1}{2} \times \frac{4}{8} \qquad \frac{2}{3} \times \frac{6}{9}$

Cross Products of a Proportion

For any proportion of the form $\frac{a}{b}=\frac{c}{d}$, where $b\neq 0, d\neq 0$, its cross products are equal.

$$a \cdot d = b \cdot c$$



Cross products can be used to test whether a proportion is true. To test whether an equation makes a proportion, we find the cross products. If they are the equal, we have a proportion.

EXAMPLE 6.41

Determine whether each equation is a proportion:

ⓐ $\frac{4}{9}=\frac{12}{28}$ ⓑ $\frac{17.5}{37.5}=\frac{7}{15}$

Solution

To determine if the equation is a proportion, we find the cross products. If they are equal, the equation is a proportion.

ⓐ

$$\frac{4}{9} = \frac{12}{28}$$

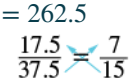
Find the cross products. $28 \cdot 4 = 112 \qquad 9 \cdot 12 = 108$

Since the cross products are not equal, $28 \cdot 4 \neq 9 \cdot 12$, the equation is not a proportion.

ⓑ

$$\frac{17.5}{37.5} = \frac{7}{15}$$

Find the cross products. $15 \cdot 17.5 = 262.5 \qquad 37.5 \cdot 7 = 262.5$



Since the cross products are equal, $15 \cdot 17.5 = 37.5 \cdot 7$, the equation is a proportion.

TRY IT : : 6.81 Determine whether each equation is a proportion:

ⓐ $\frac{7}{9}=\frac{54}{72}$ ⓑ $\frac{24.5}{45.5}=\frac{7}{13}$

TRY IT : : 6.82 Determine whether each equation is a proportion:

ⓐ $\frac{8}{9}=\frac{56}{73}$ ⓑ $\frac{28.5}{52.5}=\frac{8}{15}$

Solve Proportions

To solve a proportion containing a variable, we remember that the proportion is an equation. All of the techniques we have used so far to solve equations still apply. In the next example, we will solve a proportion by multiplying by the Least Common Denominator (LCD) using the Multiplication Property of Equality.

EXAMPLE 6.42

Solve: $\frac{x}{63} = \frac{4}{7}$.

Solution

	$\frac{x}{63} = \frac{4}{7}$
To isolate x, multiply both sides by the LCD, 63.	$63\left(\frac{x}{63}\right) = 63\left(\frac{4}{7}\right)$
Simplify.	$x = \frac{9 \cdot 7 \cdot 4}{7}$
Divide the common factors.	$x = 36$
Check: To check our answer, we substitute into the original proportion.	
	$\frac{x}{63} = \frac{4}{7}$
Substitute $x = 36$	$\frac{36}{63} \stackrel{?}{=} \frac{4}{7}$
Show common factors.	$\frac{4 \cdot 9}{7 \cdot 9} \stackrel{?}{=} \frac{4}{7}$
Simplify.	$\frac{4}{7} = \frac{4}{7}$ ✓

TRY IT : : 6.83 Solve the proportion: $\frac{n}{84} = \frac{11}{12}$.

TRY IT : : 6.84 Solve the proportion: $\frac{y}{96} = \frac{13}{12}$.

When the variable is in a denominator, we'll use the fact that the cross products of a proportion are equal to solve the proportions.

We can find the cross products of the proportion and then set them equal. Then we solve the resulting equation using our familiar techniques.

EXAMPLE 6.43

Solve: $\frac{144}{a} = \frac{9}{4}$.

Solution

Notice that the variable is in the denominator, so we will solve by finding the cross products and setting them equal.

	$\frac{144}{a} \stackrel{?}{=} \frac{9}{4}$
Find the cross products and set them equal.	$4 \cdot 144 = a \cdot 9$
Simplify.	$576 = 9a$
Divide both sides by 9.	$\frac{576}{9} = \frac{9a}{9}$
Simplify.	$64 = a$
Check your answer.	
	$\frac{144}{a} = \frac{9}{4}$
Substitute $a = 64$	$\frac{144}{64} \stackrel{?}{=} \frac{9}{4}$
Show common factors..	$\frac{9 \cdot 16}{4 \cdot 16} \stackrel{?}{=} \frac{9}{4}$
Simplify.	$\frac{9}{4} = \frac{9}{4}$ ✓

Another method to solve this would be to multiply both sides by the LCD, $4a$. Try it and verify that you get the same solution.

TRY IT : : 6.85 Solve the proportion: $\frac{91}{b} = \frac{7}{5}$.

TRY IT : : 6.86 Solve the proportion: $\frac{39}{c} = \frac{13}{8}$.

EXAMPLE 6.44

Solve: $\frac{52}{91} = \frac{-4}{y}$.

Solution

Find the cross products and set them equal.	$\frac{52}{91} = \frac{-4}{y}$
	$y \cdot 52 = 91(-4)$
Simplify.	$52y = -364$
Divide both sides by 52.	$\frac{52y}{52} = \frac{-364}{52}$
Simplify.	$y = -7$
Check:	
	$\frac{52}{91} = \frac{-4}{y}$
Substitute $y = -7$	$\frac{52}{91} \stackrel{?}{=} \frac{-4}{-7}$
Show common factors.	$\frac{13 \cdot 4}{13 \cdot 4} \stackrel{?}{=} \frac{-4}{-7}$
Simplify.	$\frac{4}{7} = \frac{4}{7}$ ✓

TRY IT : : 6.87 Solve the proportion: $\frac{84}{98} = \frac{-6}{x}$.

TRY IT : : 6.88 Solve the proportion: $\frac{-7}{y} = \frac{105}{135}$.

Solve Applications Using Proportions

The strategy for solving applications that we have used earlier in this chapter, also works for proportions, since proportions are equations. When we set up the proportion, we must make sure the units are correct—the units in the numerators match and the units in the denominators match.

EXAMPLE 6.45

When pediatricians prescribe acetaminophen to children, they prescribe 5 milliliters (ml) of acetaminophen for every 25 pounds of the child's weight. If Zoe weighs 80 pounds, how many milliliters of acetaminophen will her doctor prescribe?

Solution

Identify what you are asked to find.	How many ml of acetaminophen the doctor will prescribe
Choose a variable to represent it.	Let $a =$ ml of acetaminophen.
Write a sentence that gives the information to find it.	If 5 ml is prescribed for every 25 pounds, how much will be prescribed for 80 pounds?
Translate into a proportion.	$\frac{\text{ml}}{\text{pounds}} = \frac{\text{ml}}{\text{pounds}}$
Substitute given values—be careful of the units.	$\frac{5}{25} = \frac{a}{80}$
Multiply both sides by 80.	$80 \cdot \frac{5}{25} = 80 \cdot \frac{a}{80}$
Multiply and show common factors.	$\frac{16 \cdot 5 \cdot 5}{5 \cdot 5} = \frac{80a}{80}$
Simplify.	$16 = a$
Check if the answer is reasonable.	
Yes. Since 80 is about 3 times 25, the medicine should be about 3 times 5.	
Write a complete sentence.	The pediatrician would prescribe 16 ml of acetaminophen to Zoe.

You could also solve this proportion by setting the cross products equal.

TRY IT : : 6.89

Pediatricians prescribe 5 milliliters (ml) of acetaminophen for every 25 pounds of a child's weight. How many milliliters of acetaminophen will the doctor prescribe for Emilia, who weighs 60 pounds?

TRY IT : : 6.90

For every 1 kilogram (kg) of a child's weight, pediatricians prescribe 15 milligrams (mg) of a fever reducer. If Isabella weighs 12 kg, how many milligrams of the fever reducer will the pediatrician prescribe?

EXAMPLE 6.46

One brand of microwave popcorn has 120 calories per serving. A whole bag of this popcorn has 3.5 servings. How many calories are in a whole bag of this microwave popcorn?

Solution

Identify what you are asked to find.	How many calories are in a whole bag of microwave popcorn?
Choose a variable to represent it.	Let $c =$ number of calories.
Write a sentence that gives the information to find it.	If there are 120 calories per serving, how many calories are in a whole bag with 3.5 servings?
Translate into a proportion.	$\frac{\text{calories}}{\text{serving}} = \frac{\text{calories}}{\text{serving}}$
Substitute given values.	$\frac{120}{1} = \frac{c}{3.5}$
Multiply both sides by 3.5.	$(3.5)\left(\frac{120}{1}\right) = (3.5)\left(\frac{c}{3.5}\right)$
Multiply.	$420 = c$
Check if the answer is reasonable.	
Yes. Since 3.5 is between 3 and 4, the total calories should be between 360 (3·120) and 480 (4·120).	
Write a complete sentence.	The whole bag of microwave popcorn has 420 calories.

TRY IT : : 6.91

Marissa loves the Caramel Macchiato at the coffee shop. The 16 oz. medium size has 240 calories. How many calories will she get if she drinks the large 20 oz. size?

TRY IT : : 6.92

Yaneli loves Starburst candies, but wants to keep her snacks to 100 calories. If the candies have 160 calories for 8 pieces, how many pieces can she have in her snack?

EXAMPLE 6.47

Josiah went to Mexico for spring break and changed $\$325$ dollars into Mexican pesos. At that time, the exchange rate had $\$1$ U.S. is equal to 12.54 Mexican pesos. How many Mexican pesos did he get for his trip?

Solution

Identify what you are asked to find.	How many Mexican pesos did Josiah get?
Choose a variable to represent it.	Let $p =$ number of pesos.
Write a sentence that gives the information to find it.	If \$1 U.S. is equal to 12.54 Mexican pesos, then \$325 is how many pesos?
Translate into a proportion.	$\frac{\$}{\text{pesos}} = \frac{\$}{\text{pesos}}$
Substitute given values.	$\frac{1}{12.54} = \frac{325}{p}$
The variable is in the denominator, so find the cross products and set them equal.	$p \cdot 1 = 12.54(325)$
Simplify.	$c = 4{,}075.5$
Check if the answer is reasonable.	
Yes, \$100 would be \$1,254 pesos. \$325 is a little more than 3 times this amount.	
Write a complete sentence.	Josiah has 4075.5 pesos for his spring break trip.

TRY IT : : 6.93

Yurianna is going to Europe and wants to change $\$800$ dollars into Euros. At the current exchange rate, $\$1$ US is equal to 0.738 Euro. How many Euros will she have for her trip?

TRY IT : : 6.94

Corey and Nicole are traveling to Japan and need to exchange $\$600$ into Japanese yen. If each dollar is 94.1 yen, how many yen will they get?

Write Percent Equations As Proportions

Previously, we solved percent equations by applying the properties of equality we have used to solve equations throughout this text. Some people prefer to solve percent equations by using the proportion method. The proportion method for solving percent problems involves a percent proportion. A **percent proportion** is an equation where a percent is equal to an equivalent ratio.

For example, $60\% = \frac{60}{100}$ and we can simplify $\frac{60}{100} = \frac{3}{5}$. Since the equation $\frac{60}{100} = \frac{3}{5}$ shows a percent equal to an equivalent ratio, we call it a **percent proportion**. Using the vocabulary we used earlier:

$$\frac{\text{amount}}{\text{base}} = \frac{\text{percent}}{100}$$

$$\frac{3}{5} = \frac{60}{100}$$

Percent Proportion

The amount is to the base as the percent is to 100.

$$\frac{\text{amount}}{\text{base}} = \frac{\text{percent}}{100}$$

If we restate the problem in the words of a proportion, it may be easier to set up the proportion:

The amount is to the base as the percent is to one hundred.

We could also say:

The amount out of the base is the same as the percent out of one hundred.

First we will practice translating into a percent proportion. Later, we'll solve the proportion.

EXAMPLE 6.48

Translate to a proportion. What number is 75% of 90?

Solution

If you look for the word "of", it may help you identify the base.

Identify the parts of the percent proportion.	$\underbrace{\text{What number}}_{\text{amount}}$ is $\underbrace{75\%}_{\text{percent}}$ of $\underbrace{90?}_{\text{base}}$
Restate as a proportion.	What number out of 90 is the same as 75 out of 100?
Set up the proportion. Let $n = \text{number}$.	$\frac{n}{90} = \frac{75}{100}$

TRY IT : : 6.95 Translate to a proportion: What number is 60% of 105?

TRY IT : : 6.96 Translate to a proportion: What number is 40% of 85?

EXAMPLE 6.49

Translate to a proportion. 19 is 25% of what number?

Solution

Identify the parts of the percent proportion.	$\underbrace{19}_{\text{amount}}$ is $\underbrace{25\%}_{\text{percent}}$ of $\underbrace{\text{what number?}}_{\text{base}}$
Restate as a proportion.	19 out of what number is the same as 25 out of 100?
Set up the proportion. Let $n = \text{number}$.	$\frac{19}{n} = \frac{25}{100}$

TRY IT : : 6.97 Translate to a proportion: 36 is 25% of what number?

TRY IT : : 6.98 Translate to a proportion: 27 is 36% of what number?

EXAMPLE 6.50

Translate to a proportion. What percent of 27 is 9?

Solution

Identify the parts of the percent proportion.	$\underbrace{\text{What percent}}_{\text{percent}}\ \text{of}\ \underbrace{27}_{\text{base}}\ \text{is}\ \underbrace{9?}_{\text{amount}}$
Restate as a proportion.	9 out of 27 is the same as what number out of 100?
Set up the proportion. Let $p = \text{percent}$.	$\frac{9}{27} = \frac{p}{100}$

TRY IT : : 6.99 Translate to a proportion: What percent of 52 is 39?

TRY IT : : 6.100 Translate to a proportion: What percent of 92 is 23?

Translate and Solve Percent Proportions

Now that we have written percent equations as proportions, we are ready to solve the equations.

EXAMPLE 6.51

Translate and solve using proportions: What number is 45% of 80?

Solution

Identify the parts of the percent proportion.	$\underbrace{\text{What number}}_{\text{amount}}\ \text{is}\ \underbrace{45\%}_{\text{percent}}\ \text{of}\ \underbrace{80?}_{\text{base}}$
Restate as a proportion.	What number out of 80 is the same as 45 out of 100?
Set up the proportion. Let $n =$ number.	$\frac{n}{80} = \frac{45}{100}$
Find the cross products and set them equal.	$100 \cdot n = 80 \cdot 45$
Simplify.	$100n = 3{,}600$
Divide both sides by 100.	$\frac{100n}{100} = \frac{3{,}600}{100}$
Simplify.	$n = 36$
Check if the answer is reasonable.	
Yes. 45 is a little less than half of 100 and 36 is a little less than half 80.	
Write a complete sentence that answers the question.	36 is 45% of 80.

TRY IT : : 6.101 Translate and solve using proportions: What number is 65% of 40?

TRY IT : : 6.102 Translate and solve using proportions: What number is 85% of 40?

In the next example, the percent is more than 100, which is more than one whole. So the unknown number will be more than the base.

EXAMPLE 6.52

Translate and solve using proportions: 125% of 25 is what number?

Solution

Identify the parts of the percent proportion.	$\underbrace{125\%}_{\text{percent}}$ is $\underbrace{25}_{\text{base}}$ of $\underbrace{\text{what number?}}_{\text{amount}}$
Restate as a proportion.	What number out of 25 is the same as 125 out of 100?
Set up the proportion. Let $n =$ number.	$\frac{n}{25} = \frac{125}{100}$
Find the cross products and set them equal.	$100 \cdot n = 25 \cdot 125$
Simplify.	$100n = 3{,}125$
Divide both sides by 100.	$\frac{100n}{100} = \frac{3{,}125}{100}$
Simplify.	$n = 31.25$
Check if the answer is reasonable.	
Yes. 125 is more than 100 and 31.25 is more than 25.	
Write a complete sentence that answers the question.	125% of 25 is 31.25.

TRY IT : : 6.103 Translate and solve using proportions: 125% of 64 is what number?

TRY IT : : 6.104 Translate and solve using proportions: 175% of 84 is what number?

Percents with decimals and money are also used in proportions.

EXAMPLE 6.53

Translate and solve: 6.5% of what number is $\$1.56$?

Solution

Identify the parts of the percent proportion.	$\underbrace{6.5\%}_{\text{percent}}$ of $\underbrace{\text{what number}}_{\text{base}}$ is $\underbrace{\$1.56?}_{\text{amount}}$
Restate as a proportion.	\$1.56 out of what number is the same as 6.5 out of 100?
Set up the proportion. Let $n =$ number.	$\frac{1.56}{n} = \frac{6.5}{100}$
Find the cross products and set them equal.	$100(1.56) = n \cdot 6.5$
Simplify.	$156 = 6.5n$
Divide both sides by 6.5 to isolate the variable.	$\frac{156}{6.5} = \frac{6.5n}{6.5}$
Simplify.	$24 = n$
Check if the answer is reasonable.	
Yes. 6.5% is a small amount and \$1.56 is much less than \$24.	
Write a complete sentence that answers the question.	6.5% of \$24 is \$1.56.

TRY IT : : 6.105 Translate and solve using proportions: 8.5% of what number is $\$3.23$?

TRY IT : : 6.106 Translate and solve using proportions: 7.25% of what number is $\$4.64$?

EXAMPLE 6.54

Translate and solve using proportions: What percent of 72 is 9?

Solution

Identify the parts of the percent proportion.	$\underbrace{\text{What percent}}_{\text{percent}}\ \text{of}\ \underbrace{72}_{\text{base}}\ \text{is}\ \underbrace{9?}_{\text{amount}}$
Restate as a proportion.	9 out of 72 is the same as what number out of 100?
Set up the proportion. Let $n =$ number.	$\frac{9}{72} = \frac{n}{100}$
Find the cross products and set them equal.	$72 \cdot n = 100 \cdot 9$
Simplify.	$72n = 900$
Divide both sides by 72.	$\frac{72n}{72} = \frac{900}{72}$
Simplify.	$n = 12.5$
Check if the answer is reasonable.	
Yes. 9 is $\frac{1}{8}$ of 72 and $\frac{1}{8}$ is 12.5%.	
Write a complete sentence that answers the question.	12.5% of 72 is 9.

TRY IT : : 6.107 Translate and solve using proportions: What percent of 72 is 27?

TRY IT : : 6.108 Translate and solve using proportions: What percent of 92 is 23?

6.5 EXERCISES

Practice Makes Perfect

Use the Definition of Proportion

In the following exercises, write each sentence as a proportion.

244. 4 is to 15 as 36 is to 135.

245. 7 is to 9 as 35 is to 45.

246. 12 is to 5 as 96 is to 40.

247. 15 is to 8 as 75 is to 40.

248. 5 wins in 7 games is the same as 115 wins in 161 games.

249. 4 wins in 9 games is the same as 36 wins in 81 games.

250. 8 campers to 1 counselor is the same as 48 campers to 6 counselors.

251. 6 campers to 1 counselor is the same as 48 campers to 8 counselors.

252. \$9.36 for 18 ounces is the same as \$2.60 for 5 ounces.

253. \$3.92 for 8 ounces is the same as \$1.47 for 3 ounces.

254. \$18.04 for 11 pounds is the same as \$4.92 for 3 pounds.

255. \$12.42 for 27 pounds is the same as \$5.52 for 12 pounds.

In the following exercises, determine whether each equation is a proportion.

256. $\frac{7}{15} = \frac{56}{120}$

257. $\frac{5}{12} = \frac{45}{108}$

258. $\frac{11}{6} = \frac{21}{16}$

259. $\frac{9}{4} = \frac{39}{34}$

260. $\frac{12}{18} = \frac{4.99}{7.56}$

261. $\frac{9}{16} = \frac{2.16}{3.89}$

262. $\frac{13.5}{8.5} = \frac{31.05}{19.55}$

263. $\frac{10.1}{8.4} = \frac{3.03}{2.52}$

Solve Proportions

In the following exercises, solve each proportion.

264. $\frac{x}{56} = \frac{7}{8}$

265. $\frac{n}{91} = \frac{8}{13}$

266. $\frac{49}{63} = \frac{z}{9}$

267. $\frac{56}{72} = \frac{y}{9}$

268. $\frac{5}{a} = \frac{65}{117}$

269. $\frac{4}{b} = \frac{64}{144}$

270. $\frac{98}{154} = \frac{-7}{p}$

271. $\frac{72}{156} = \frac{-6}{q}$

272. $\frac{a}{-8} = \frac{-42}{48}$

273. $\frac{b}{-7} = \frac{-30}{42}$

274. $\frac{2.6}{3.9} = \frac{c}{3}$

275. $\frac{2.7}{3.6} = \frac{d}{4}$

276. $\frac{2.7}{j} = \frac{0.9}{0.2}$

277. $\frac{2.8}{k} = \frac{2.1}{1.5}$

278. $\frac{\frac{1}{2}}{1} = \frac{m}{8}$

279. $\frac{\frac{1}{3}}{3} = \frac{9}{n}$

Solve Applications Using Proportions

In the following exercises, solve the proportion problem.

280. Pediatricians prescribe 5 milliliters (ml) of acetaminophen for every 25 pounds of a child's weight. How many milliliters of acetaminophen will the doctor prescribe for Jocelyn, who weighs 45 pounds?

281. Brianna, who weighs 6 kg, just received her shots and needs a pain killer. The pain killer is prescribed for children at 15 milligrams (mg) for every 1 kilogram (kg) of the child's weight. How many milligrams will the doctor prescribe?

282. At the gym, Carol takes her pulse for 10 sec and counts 19 beats. How many beats per minute is this? Has Carol met her target heart rate of 140 beats per minute?

283. Kevin wants to keep his heart rate at 160 beats per minute while training. During his workout he counts 27 beats in 10 seconds. How many beats per minute is this? Has Kevin met his target heart rate?

284. A new energy drink advertises 106 calories for 8 ounces. How many calories are in 12 ounces of the drink?

285. One 12 ounce can of soda has 150 calories. If Josiah drinks the big 32 ounce size from the local mini-mart, how many calories does he get?

286. Karen eats $\frac{1}{2}$ cup of oatmeal that counts for 2 points on her weight loss program. Her husband, Joe, can have 3 points of oatmeal for breakfast. How much oatmeal can he have?

287. An oatmeal cookie recipe calls for $\frac{1}{2}$ cup of butter to make 4 dozen cookies. Hilda needs to make 10 dozen cookies for the bake sale. How many cups of butter will she need?

288. Janice is traveling to Canada and will change $\$250$ US dollars into Canadian dollars. At the current exchange rate, $\$1$ US is equal to $\$1.01$ Canadian. How many Canadian dollars will she get for her trip?

289. Todd is traveling to Mexico and needs to exchange $\$450$ into Mexican pesos. If each dollar is worth 12.29 pesos, how many pesos will he get for his trip?

290. Steve changed $\$600$ into 480 Euros. How many Euros did he receive per US dollar?

291. Martha changed $\$350$ US into 385 Australian dollars. How many Australian dollars did she receive per US dollar?

292. At the laundromat, Lucy changed $\$12.00$ into quarters. How many quarters did she get?

293. When she arrived at a casino, Gerty changed $\$20$ into nickels. How many nickels did she get?

294. Jesse's car gets 30 miles per gallon of gas. If Las Vegas is 285 miles away, how many gallons of gas are needed to get there and then home? If gas is $\$3.09$ per gallon, what is the total cost of the gas for the trip?

295. Danny wants to drive to Phoenix to see his grandfather. Phoenix is 370 miles from Danny's home and his car gets 18.5 miles per gallon. How many gallons of gas will Danny need to get to and from Phoenix? If gas is $\$3.19$ per gallon, what is the total cost for the gas to drive to see his grandfather?

296. Hugh leaves early one morning to drive from his home in Chicago to go to Mount Rushmore, 812 miles away. After 3 hours, he has gone 190 miles. At that rate, how long will the whole drive take?

297. Kelly leaves her home in Seattle to drive to Spokane, a distance of 280 miles. After 2 hours, she has gone 152 miles. At that rate, how long will the whole drive take?

298. Phil wants to fertilize his lawn. Each bag of fertilizer covers about $4{,}000$ square feet of lawn. Phil's lawn is approximately $13{,}500$ square feet. How many bags of fertilizer will he have to buy?

299. April wants to paint the exterior of her house. One gallon of paint covers about 350 square feet, and the exterior of the house measures approximately 2000 square feet. How many gallons of paint will she have to buy?

Write Percent Equations as Proportions

In the following exercises, translate to a proportion.

300. What number is 35% of 250?

301. What number is 75% of 920?

302. What number is 110% of 47?

303. What number is 150% of 64?

304. 45 is 30% of what number?

305. 25 is 80% of what number?

306. 90 is 150% of what number?

307. 77 is 110% of what number?

308. What percent of 85 is 17?

309. What percent of 92 is 46?

310. What percent of 260 is 340?

311. What percent of 180 is 220?

Translate and Solve Percent Proportions

In the following exercises, translate and solve using proportions.

312. What number is 65% of 180?

313. What number is 55% of 300?

314. 18% of 92 is what number?

315. 22% of 74 is what number?

316. 175% of 26 is what number?

317. 250% of 61 is what number?

318. What is 300% of 488?

319. What is 500% of 315?

320. 17% of what number is $\$7.65$?

321. 19% of what number is $\$6.46$?

322. $\$13.53$ is 8.25% of what number?

323. $\$18.12$ is 7.55% of what number?

324. What percent of 56 is 14?

325. What percent of 80 is 28?

326. What percent of 96 is 12?

327. What percent of 120 is 27?

Everyday Math

328. Mixing a concentrate Sam bought a large bottle of concentrated cleaning solution at the warehouse store. He must mix the concentrate with water to make a solution for washing his windows. The directions tell him to mix 3 ounces of concentrate with 5 ounces of water. If he puts 12 ounces of concentrate in a bucket, how many ounces of water should he add? How many ounces of the solution will he have altogether?

329. Mixing a concentrate Travis is going to wash his car. The directions on the bottle of car wash concentrate say to mix 2 ounces of concentrate with 15 ounces of water. If Travis puts 6 ounces of concentrate in a bucket, how much water must he mix with the concentrate?

Writing Exercises

330. To solve "what number is 45% of 350" do you prefer to use an equation like you did in the section on Decimal Operations or a proportion like you did in this section? Explain your reason.

331. To solve "what percent of 125 is 25" do you prefer to use an equation like you did in the section on Decimal Operations or a proportion like you did in this section? Explain your reason.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
use definition of proportion.			
solve proportions.			
solve applications using proportions.			
write percent equations as proportions.			
translate and solve percent proportions.			

ⓑ *Overall, after looking at the checklist, do you think you are well-prepared for the next Chapter? Why or why not?*